

Yuan (Paul) Zhuang

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Education

University of Connecticut, Ph.D., Computer Science and Engineering, Storrs, CT	Sept 2024 – Present
University of Pennsylvania, M.S. Data Science, Philadelphia, PA (GPA: 3.87/4.00)	May 2024
Zhejiang University, B.Eng. Ocean Engineering and Technology, Hangzhou, China GPA: (3.65/4.00)	Jun 2022

Internship

Research Intern, Microsoft, Microsoft Research Lab – Redmond, WA, US	May 2026 – Aug 2026
• Large Language Models' architecture design for Mixture-of-Experts and Large Language Model test time training	
Research Intern, JING DONG, Embodied Technology Group, Beijing, China	May 2025 – Aug 2025
• Fine-tuned Vision Language Action (VLA) models for parcel sorting and palletizing, enhancing automation in logistics	
• Investigated and developed novel model architectures to improve VLA performance on multi-task and unseen domains	

Research

Low-Rank Adaptation for Critic Learning in Off-Policy Reinforcement Learning, First Coauthor <i>University of Connecticut, Miao Group</i>	Nov 2025 – Present
• Proposed a Low-Rank Adaptation regularization method for critic to boost the performance of reinforcement learning	
• Currently under review: Arxiv Link and the Github repository: Github link	
Masked Vision Language Action Models(VLA), Coauthor <i>University of Connecticut, Miao Group</i>	Oct 2025 – present
• Proposed a masked embedding method to enhance the performance and robustness of VLA models	
• Reduced the influence from the background to avoid VLA performance degrade due to the changes of environments	
LD-MoLE: Learnable Dynamic Routing for Mixture of LoRA Experts, First Coauthor <i>University of Connecticut, Miao Group</i>	Jan 2025 – Oct 2025
• Proposed a novel, fully differentiable and dynamic routing mechanism to learn experts allocation for Mixture of Experts	
• Introduced an analytical sparsity control objective to regularize the number of activated experts	
• Achieved superior performance on common benchmarks while maintaining model sparsity	
• Published in ICLR 2026 : Arxiv Link and the Github repository: Github link	
YOLO-MARL: You Only LLM Once for Multi-agent Reinforcement Learning, First Coauthor <i>University of Connecticut, Miao Group</i>	May 2024 – Sept 2024
• Proposed a framework to synergize the planning capabilities of LLMs with Multi-Agent Reinforcement Learning	
• Designed a zero-shot prompting approach for Python code generation for various game environments with minimal user input	
• Published in IROS 2025 : Arxiv Link and the Github repository: Github link	
Swarm Control based on Graph Neural Networks, Research Assistant <i>University of Pennsylvania, ScalAR Lab @GRASP</i>	May 2023 – Sept 2023
• Constructed a graph neural network model to reconstruct swarm relationships and identify the leader within the group	

Projects

Image to Image Translation: Trade off between diversity and fidelity, Advanced Topics In Machine Perception
• Implemented BicycleGAN and designed a new Soft-IntroVAE architecture to generate higher quality images: Github link
Neural Style Transfer, Computer Vision & Computational Photography
• Applied different neural networks to obtained stylized video: Origin video , Leon's NST video , Johnson's NST video

Awards & Honors

University of Connecticut	2024 – Present
• Pratt & Whitney Advanced Systems Engineering Fellowship (2024, 2025)	
Zhejiang University	2018 – 2022
• Outstanding Graduates(2022), University Scholarship(2018, 2019)	
• Second Prize at the 2nd National Underwater Robotics Competition(2021)	